Is the leak-free network running DD?

DD = SINGLE-CANDIDATE ELIMINATION, sudoku's 'naked single', on a graph (next slide). Causal interventions, not probe scores.

YES -- it applies DD's elimination rule, and chains it off its own deductions.
The withhold test is passed by ALL THREE models, so it is not evidence on its own.
The ORDER it emits them in is a fixed index sweep, not a deduction.

THREE 12-LAYER TRANSFORMERS, trained on Direct Deduction's own solution traces.

SOLVER	leak-free encoding, 240,000 training instances. Solves 100.0% of held-out graphs.
WEAK TWIN	leak-free encoding, 80,000 instances -- the only difference. Solves 85.4%.
LEAK-READER	LEAKED encoding, 80,000 instances. Solves 99.6% by reading the answer off the input.

ONE-TOKEN COUNTERFACTUALS: change one input token so DD's own answer changes, and ask whether the model's answer follows. Each cell = the fraction of cases where it did what DD does, so HIGH = followed DD.

	RULE on given hints	RULE on its OWN facts	WITHHOLD forcing broken	placebo edit still names v
SOLVER	1.000	0.980	1.000	0.940
WEAK TWIN	1.000	0.980	0.824	0.900
LEAK-READER	0.388	0.008	0.992	0.964
chance	0.250	0.250	--	--

WHY INTERVENTIONS AND NOT PROBES: the best single feature for this model's own 'which node is next' scores 0.816 on a decoding metric (max-over-nodes F1) -- and a hand rule that deduces nothing at all, 'the next unknown index after the last one', already scores 0.632 on that same metric. A decoding score cannot separate deducing from counting. Editing one token can.

CAVEAT, up front: one task configuration (N = 24, degree 5, q = 4) and ONE CHECKPOINT PER MODEL, no seed replicates -- so every leak-free vs leaked contrast is also two separate training runs. And YES means DD's PROPAGATION STEP: these tests do not separate DD from another propagator with the same one-step behaviour (last slide).

The task, and the algorithm the model imitates

THE TASK

Colour a random regular graph: N = 24 nodes, every node of degree 5, using q = 4 colours so that no edge joins two nodes of the same colour.

Some nodes are given as HINTS (15 of 24 here); the rest must be deduced.

Each instance is built from a planted colouring by removing hints one at a time, keeping a removal only if DD can still recover everything. So every instance is solvable by DD alone, and instances differ only in how deep the deduction has to go.

DD -- DIRECT DEDUCTION

SINGLE-CANDIDATE ELIMINATION (sudoku's 'naked single'), on a graph: a node becomes determined one round after its known neighbours show all q-1 = 3 colours other than its own. Sweep the nodes, emit every determined one, repeat until none are left.

THE FORCING SET of a node v = the known neighbours whose colours do the eliminating. DD fires on v exactly when that set shows q-1 = 3 distinct colours; v then takes the one colour left over.

THE BINDING NEIGHBOUR of v = the one member of that set whose colour rules out v's last alternative; its fellow members are just as known, but not decisive. Every intervention in this deck edits the binding neighbour.

REASONING DEPTH T = how many rounds an instance needs, 1 to 6 at this hint count (15 of 24). The example on the next slide needs 2.

Solvable-by-DD is a property of the INSTANCES, not a proof that no other method works -- which is why the deck goes on to rule out the obvious non-deductive routes (reading the leak, next slide; emitting nodes by index order, the last slide) rather than assuming they are closed.

The leak, and why there are two kinds of model

THE MODEL

A 12-layer decoder-only transformer (d_model 256, 4 heads per layer, MLP 1024), trained from scratch with next-token cross-entropy on DD's own traces: graph, hints, then round by round the nodes DD determines and the colours it gives them.

SOLVE RATE, which also names each model, is scored by FREE GENERATION: the fraction of 1500 held-out instances it colours completely and correctly with no teacher forcing.

THE LEAK

The generator listed each node's neighbours sorted by the neighbour's TRUE COLOUR. So a neighbour's POSITION in the adjacency row encoded its colour, and 61-67% of nodes could be read off with no deduction at all. Permuting the rows at test time collapses a model trained on them: 0.9975 -> 0.1450 (n=400 order ablation) -- load-bearing, not incidental.

LEAK-READER	sorted rows, 80,000 instances. Solves 99.6%. follows the elimination rule on 0.388 of counterfactuals vs 0.250 chance (slide 8); how it still reaches 99.6% is not measured.
SOLVER	rows permuted, 240,000 instances. Solves 100.0%.
WEAK TWIN	rows permuted, 80,000 instances -- the ONLY difference from SOLVER. Solves 85.4%.

THE QUESTION: the SOLVER colours a task whose instances are ALL solvable by DD. Is it running DD, or something else that happens to work here?

Every BEHAVIOURAL number from here on is causal: edit one input token, ask whether the output follows DD's answer to the EDITED problem. (The one correlational page is labelled as such.)

THE OTHER NON-DEDUCTIVE ROUTE -- INDEX ORDER. DD's traces emit each round's nodes in ASCENDING NODE INDEX in 100.0% of rounds (3,746 rounds, 1,000 instances), so the ORDER needs no deduction at all. The model's best 'which node is next' feature scores 0.816 on a decoding metric (max-over-nodes F1) -- but a hand rule that deduces nothing, 'the next unknown index after the last one', already scores 0.632 on it. Most of that score is counting. Only WHICH nodes get emitted can carry deduction -- what the withhold test measures.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-33	`N0 N1 ... N23`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 198 TOKENS

HINTS	`BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)`
GRAPH	`BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 2 rounds, 9 deductions,
18 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=COLOR_2 N3=COLOR_2 N5=COLOR_2 N8=COLOR_3 N10=COLOR_0 N11=COLOR_1 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND          N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours N16 N5 N18 N13 N14

their colours known before this round:

```
[0, 2, 3]  (3 = q-1 of the 4)

(4 known neighbours, 3 DISTINCT colours -- some repeat)

still unknown: N16
```

only colour left: 1

```
so the trace emits  N0 COLOR_1
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The intervention edits ONE colour token: in this section (the CHAINED EDIT) or back in the hints (the HINT EDIT).

T=2 rounds, the median of 40 draws from this config (range 1-6) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The intervention: one token, two different correct answers

```
DD forces node v because v's FORCING SET -- its known neighbours -- already uses q-1 = 3
of the 4 colours. Exactly one of them, the neighbour w, holds colour a. So v must take
the one colour left over: c. Recolour w to c, and now a is the colour nobody holds, so
DD's answer for v changes from c to a.

   clean     ... BEGIN_ROUND  v  ->  [c]        w holds a, so c is v's only option
   corrupt   ... BEGIN_ROUND  v  ->  [a]        w recoloured to c, so a is now free

The ONLY difference between the two prompts is w's single colour token, and the code
checks that mechanically on every case rather than assuming it. Scored as 'moved to
DD's new answer' -- did the prediction at v become a? Not 'is it still accurate': the
correct answer CHANGES, so a memorised output scores zero. The model must pick one of
q = 4 colours, so chance on this test is 0.25.

Two versions, differing only in WHERE the edited token sits:

   HINT EDIT     w is a HINT, handed to the model in the prompt, so the edit lands
                 before any reasoning. Tests the elimination rule on GIVEN facts.

   CHAINED EDIT  w was DEDUCED by the model itself in an earlier round, so the edited
                 token sits in the REASONING section. Tests elimination against a fact
                 the model produced -- propagation, not one rule application.
```

One weak direction, stated up front: editing an emitted deduction makes the prefix inconsistent, so a model that DETECTS the edit and keeps its own answer scores as 'does not chain'. Those cases are counted separately, as 'kept its old answer'.

The converse intervention: break the forcing, and v should vanish

Both tests on the previous page EDIT the forcing set of a node the prefix has named, and score the COLOUR the model writes. They hold the NODE FIXED, so they say nothing about the other half of DD: WHICH node the model names as determined. So break the forcing instead of redirecting it -- the WITHHOLD test:

   recolour w to a colour that ANOTHER known neighbour of v already holds.

Now the freed colour a is held by nobody, v's forcing set covers only q-2 = 2 colours, v has TWO candidates, and DD does not emit v in this round at all.

The measurement moves ONE TOKEN TO THE LEFT: from the logits that predict v's COLOUR to the logits that predict WHICH NODE comes next. Does the model stop naming v?

```
   clean      ... BEGIN_ROUND  [v]  ->  c       DD emits v, and so should the model
   broken     ... BEGIN_ROUND  [ ? ]           v is no longer determined
```

PLACEBO EDIT, run on every single case: the same kind of one-token edit, but on a hint that is NOT a neighbour of v. v stays forced and DD's round is unchanged, so the model should still name v. That is the CONTROL the broken rate is read against -- without it, a low rate would only show that edits perturb the node stream. Reported the same way round as every other number in this deck: how often the model did what DD does, so HIGH = followed DD.

Two caveats, both stated before the result. (1) If the placebo collapsed too, the test would say nothing. It does not collapse, but it does not reach 1.000 either (0.90-0.96 across models), so a few points of every rate are edit perturbation -- which is why all three models are read against it. (2) ALL THREE MODELS PASS THIS TEST, the leaked one included. Detecting that a node stopped being determined is evidently easier than applying the rule, so the withhold test is not evidence of DD on its own. Its job is to separate the two leak-free twins.

Every causal measurement

model	test	n	clean	followed DD (95% Wilson)	kept its old answer	control	verdict
SOLVER	hint edit	350	1.000	**1.000 [0.99, 1.00]**	--	chance 0.250	uses the rule
SOLVER	chained edit	350	1.000	**0.980 [0.96, 0.99]**	0.020	chance 0.250	chains
SOLVER	withhold	350	1.000	**1.000 [0.99, 1.00]**	0.000	placebo 0.940 p = 2e-99	withholds
WEAK TWIN	hint edit	250	1.000	**1.000 [0.98, 1.00]**	--	chance 0.250	uses the rule
WEAK TWIN	chained edit	250	1.000	**0.980 [0.95, 0.99]**	0.008	chance 0.250	chains
WEAK TWIN	withhold	250	0.936	**0.824 [0.77, 0.87]**	0.176	placebo 0.900 p = 1e-50	withholds
LEAK-READER	hint edit	250	1.000	**0.388 [0.33, 0.45]**	--	chance 0.250	above chance, barely
LEAK-READER	chained edit	250	1.000	**0.008 [0.00, 0.03]**	0.992	chance 0.250	keeps its own answer
LEAK-READER	withhold	250	1.000	**0.992 [0.97, 1.00]**	0.008	placebo 0.964 p = 2e-72	withholds

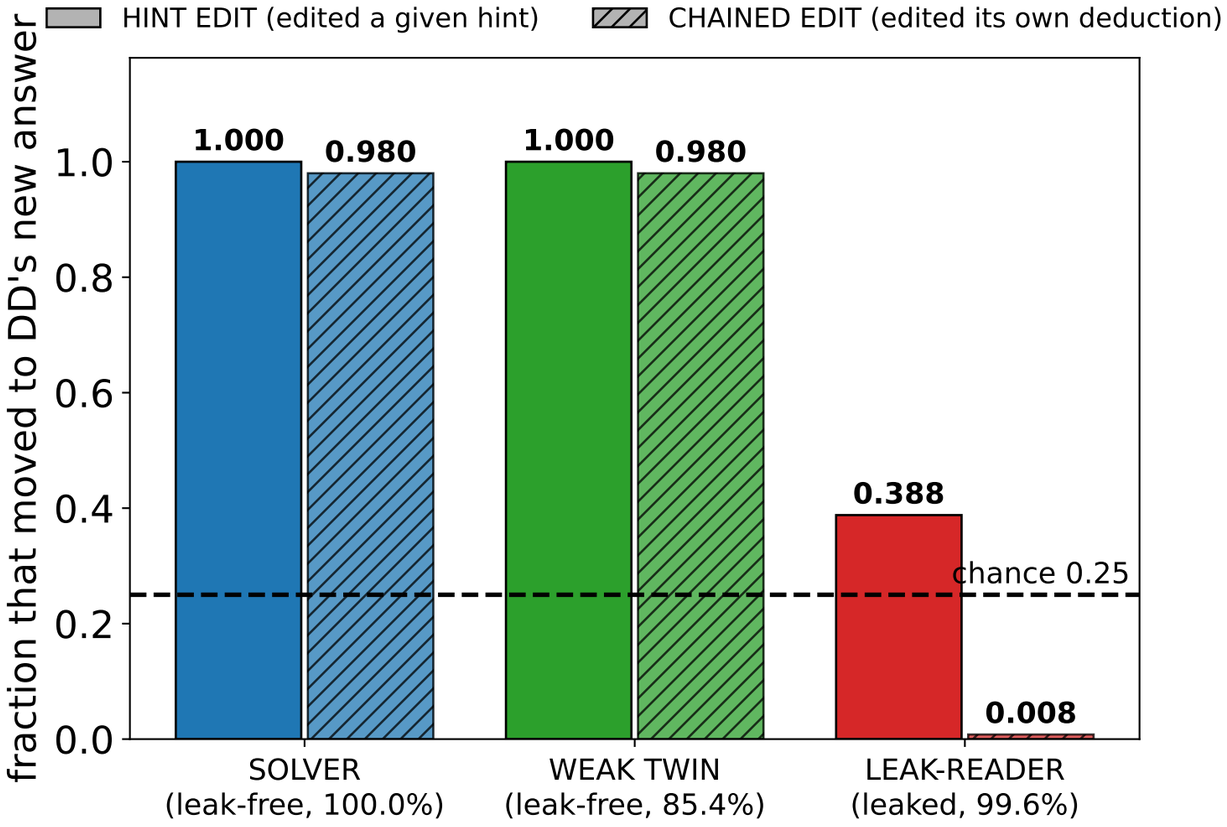
Every column is scored so that HIGH = followed DD, the placebo included; what 'followed DD' MEANS is set by the test -- emitting DD's new colour on the two rule rows, withholding the node on the withhold row. A low 'followed DD' next to a high 'clean' means the model is right for some reason other than the elimination: exactly LEAK-READER's profile on its two rule rows.

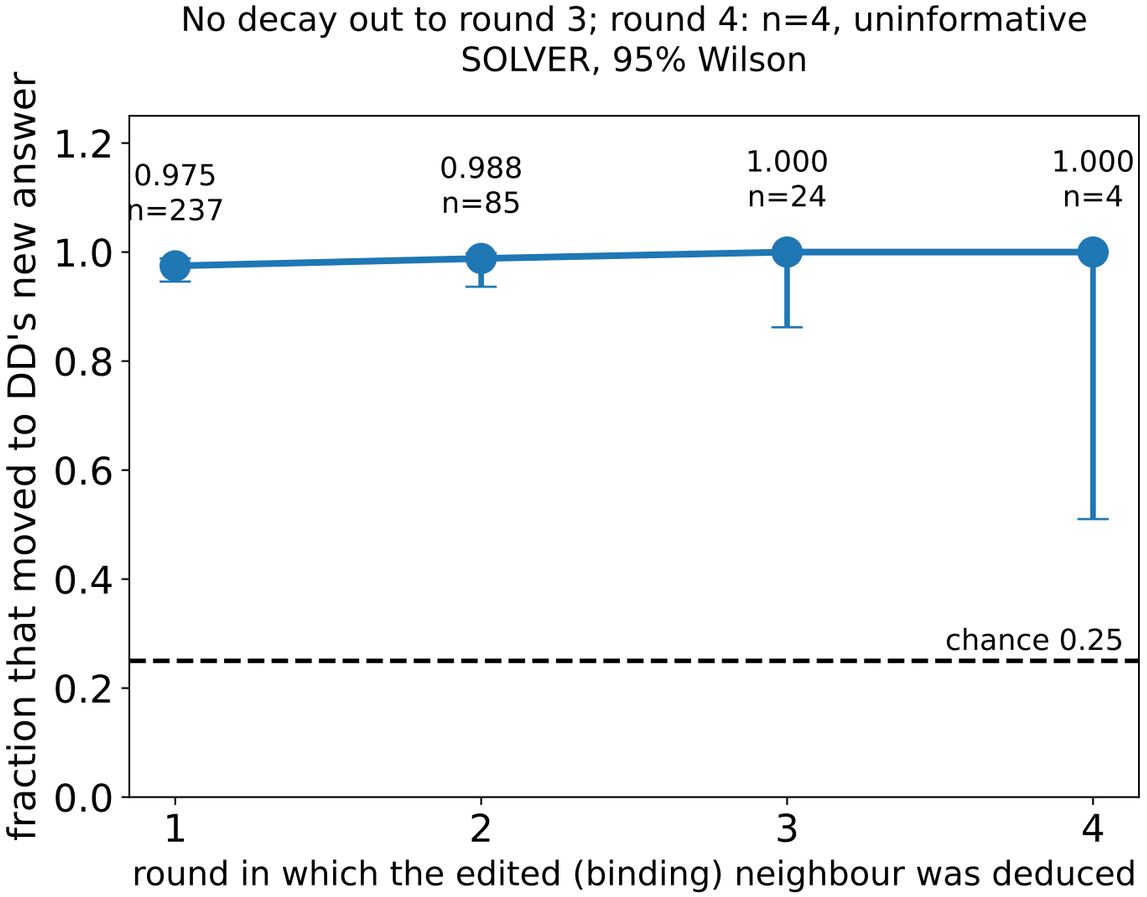
`LOGIT GAP on the hint-edit rows, logit(DD's edited answer) - logit(DD's clean answer) at the position that predicts v's colour:`

`SOLVER -15.5 -> +15.7   WEAK TWIN -15.0 -> +14.9   LEAK-READER -14.5 -> -2.0`

'clean' = the same case with no edit. 95% WILSON = the 95% interval for a proportion; EXACT McNEMAR = the paired test that counts only the cases where two conditions disagree. A sign change in the logit gap is a reversal of preference, not a shift in confidence. 'kept its old answer' = named the node anyway (withhold) or re-asserted its own colour (chained). 'control' is chance, 1/q, on the rule rows; on the withhold row the placebo edit -- the same edit on a hint that is NOT a neighbour of v, so DD's round is unchanged -- and p pairs the two WITHHOLD rates, same cases: broken against 1 - the placebo shown. n is 350 or 250 by when each run was submitted, not by filtering; paired tests use the 250 shared cases, at most 2 per instance.

The rule holds on given facts, and on facts it derived itself

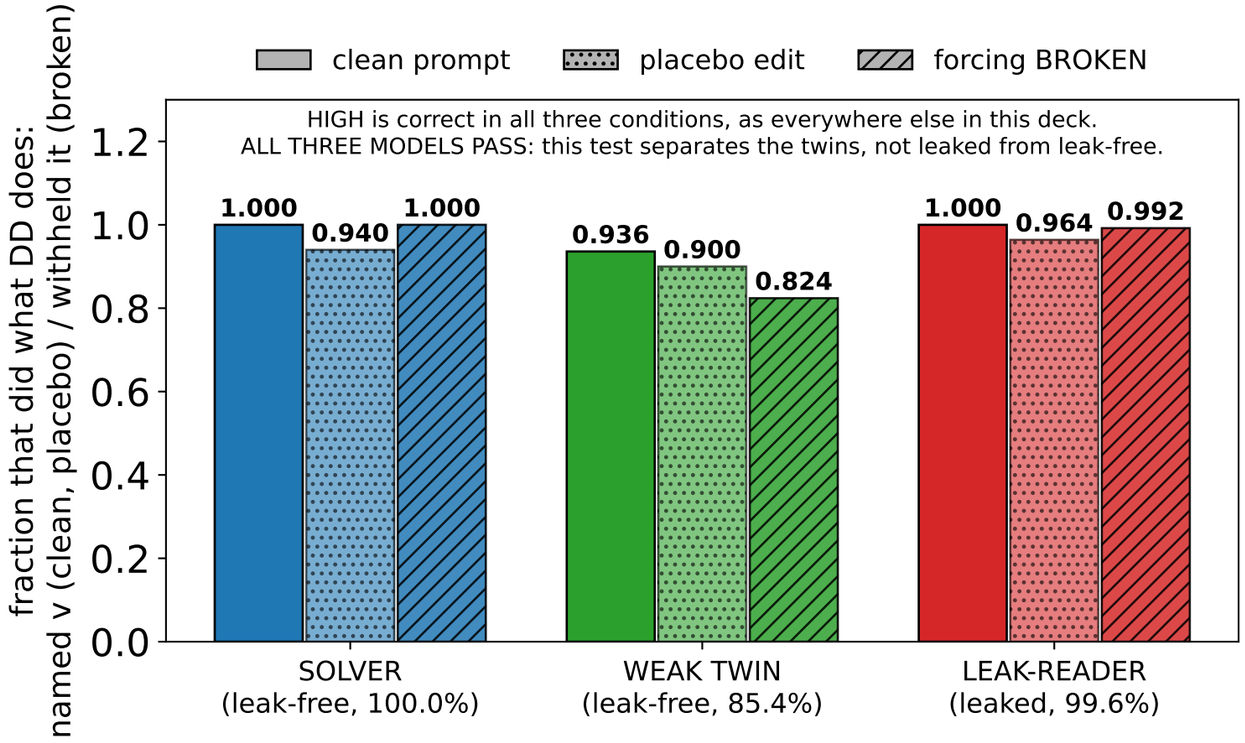


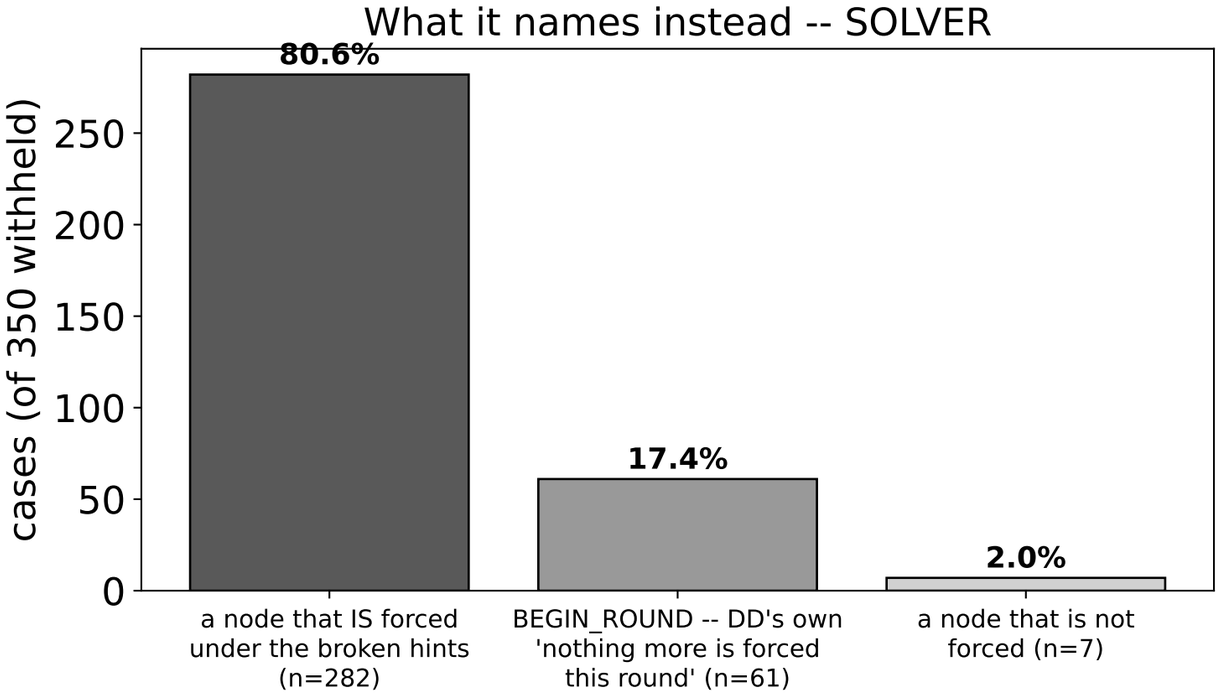


If the model chained one step but not three, round 1 would be high and the later rounds would fall off.
Through round 3 the rate never falls below 0.975, with every interval clear
of chance. Round 4 (n=4) is shown for completeness only.

Right panel splits the chained test by how deep the edited deduction sat -- the round in which the BINDING NEIGHBOUR was deduced, not the round of the node under test. Deep bins are small because deep chained forcings are rare in these instances, not because anything was filtered. Rounds 5-6 are absent because no case in the sample had its binding neighbour deduced that late.

Break the forcing and the SOLVER does not just go quiet: it names whatever else DD would name (80.6%), or ends the round (17.4%). Only 2.0% name an unforced node.

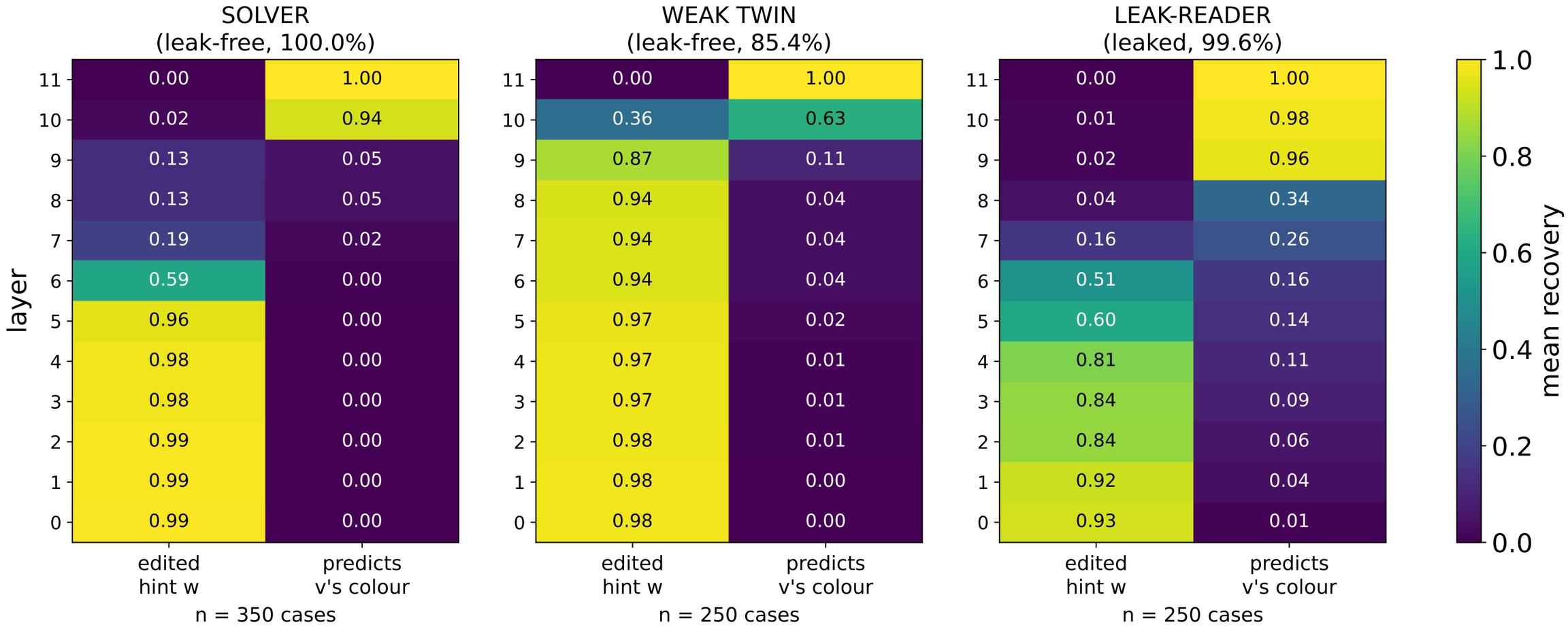




Against its WEAK TWIN the discordance is entirely one-sided: of 250 paired cases, 44 where the twin kept naming v and the SOLVER withheld it, 0 the other way (exact McNemar p = 1e-13). ONE SEED PER ARM, so that is a within-pair consistency result, not a measured effect of training-set size.

Same cases as the hint-edit test, and the case lists are byte-identical across models, so the arms are paired instance for instance. 0.176 = the WEAK TWIN still naming v with the forcing broken; on just the cases it gets right on the clean prompt it still names v on 0.188 (n=234), so that is not only the twin being wrong more often. Its 0.176 withhold gap is the same order as its 0.146 solve-rate gap -- different measurements on different samples, so a consistency check, not a decomposition. The LEAK-READER withholds too (0.992), which is why this page is a twin comparison and not evidence for DD on its own: a model can stop answering for reasons that have nothing to do with elimination.

Where the recoloured fact travels (activation patching)



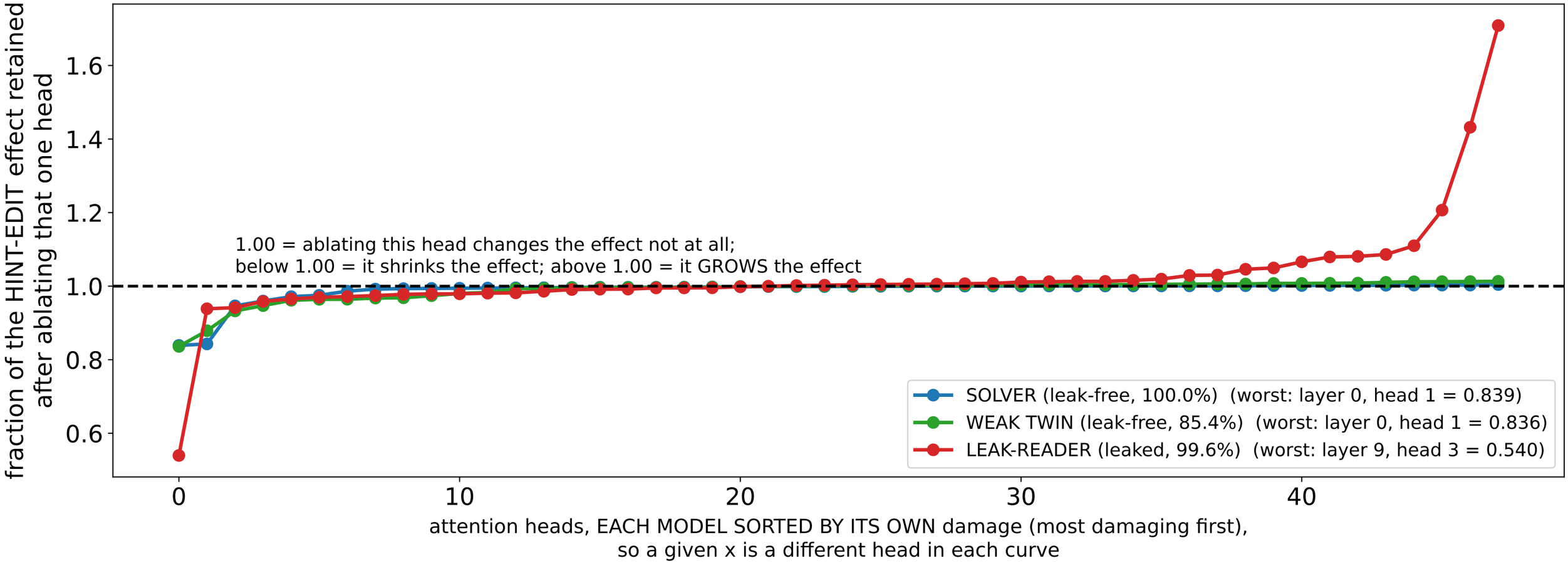
n = 350 cases n = 250 cases n = 250 cases

RECOVERY = copy one activation from the clean run into the corrupted run at that (layer, position) and measure how much of the flip it undoes: 1.00 = that one activation carries the whole edited fact, 0.00 = none of it. w is the neighbour whose colour the edit changes.

The two CONTROL columns -- the adjacency rows of w and of another known neighbour u -- never exceed 0.01 anywhere, so the EDITED FACT does not travel through the adjacency rows; they are left out of the grid to keep it readable.

Read upwards: the edited colour is held AT the edited hint through L5, handed off over L6-L10, and is at the prediction position by L10 -- one layer before the output. L11 is the last layer, so patching it there MUST restore the edit: that is the built-in positive control, and it reads 1.00. Descriptively, the SOLVER releases the fact from the hint token about 3 layers earlier than the WEAK TWIN (crossing half recovery at L6.2 against L9.7), both delivering it by L10. One seed per arm, so read that as a difference between these two checkpoints, not as a depth-of-handoff effect. In the LEAK-READER's panel recovery is a fraction of an effect that is near chance to begin with (it follows the edit on 0.388 of cases), so its numbers localise a weak sensitivity, not a rule.

No single head carries the rule: removing the worst one leaves 84% of it

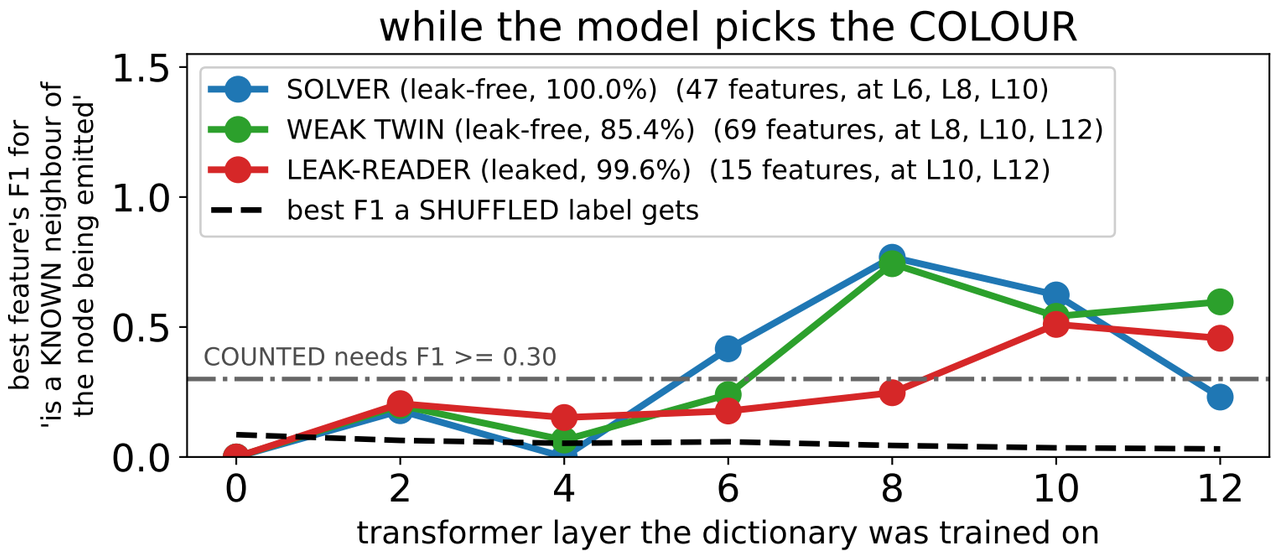


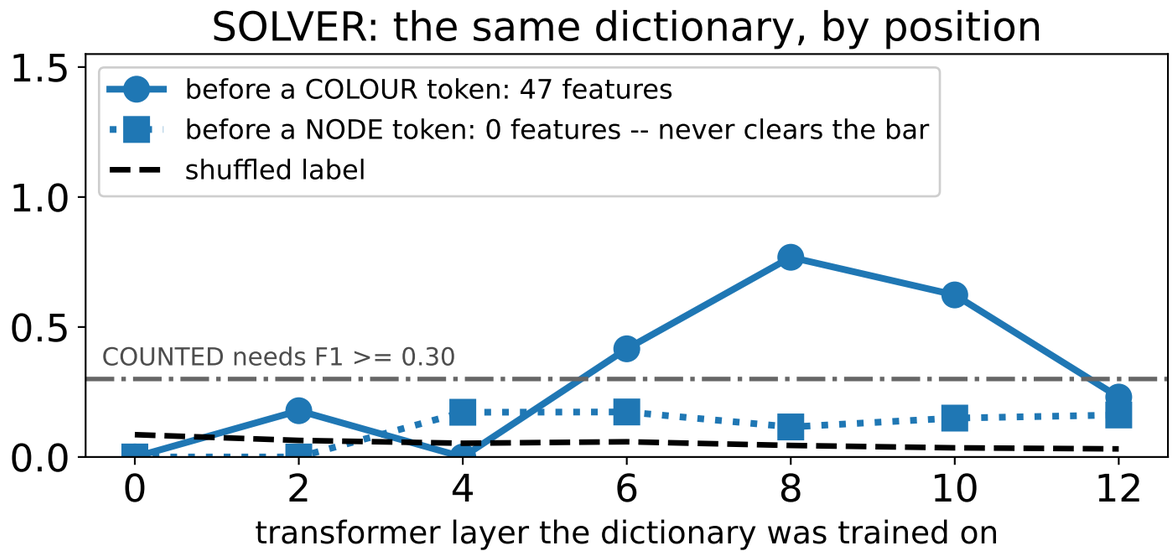
THE EFFECT is slide 8's hint-edit result: how often the model moves to DD's new answer after one token is changed.

The LEAK-READER's curve only looks steeper because its effect is near chance to begin with: a fraction of almost nothing.

Each point is one of the 48 heads (12 layers x 4 heads), zeroed one at a time, averaged over the same 350 (SOLVER) or 250 (the other two) hint-edit cases as the rule test. Values above 1.00 mean ablating that head makes the model follow the edit MORE, which happens when a head is pulling toward the pre-edit answer. So there is no single circuit to point at: the rule is distributed across heads, and this is a negative result about localisation, not about DD.

ALL THREE models carry sparse features for DD's own forcing set. In the SOLVER they peak at L8, inside the L6-L10 window where the causal patch hands the fact off -- a coincidence of location, not a second result





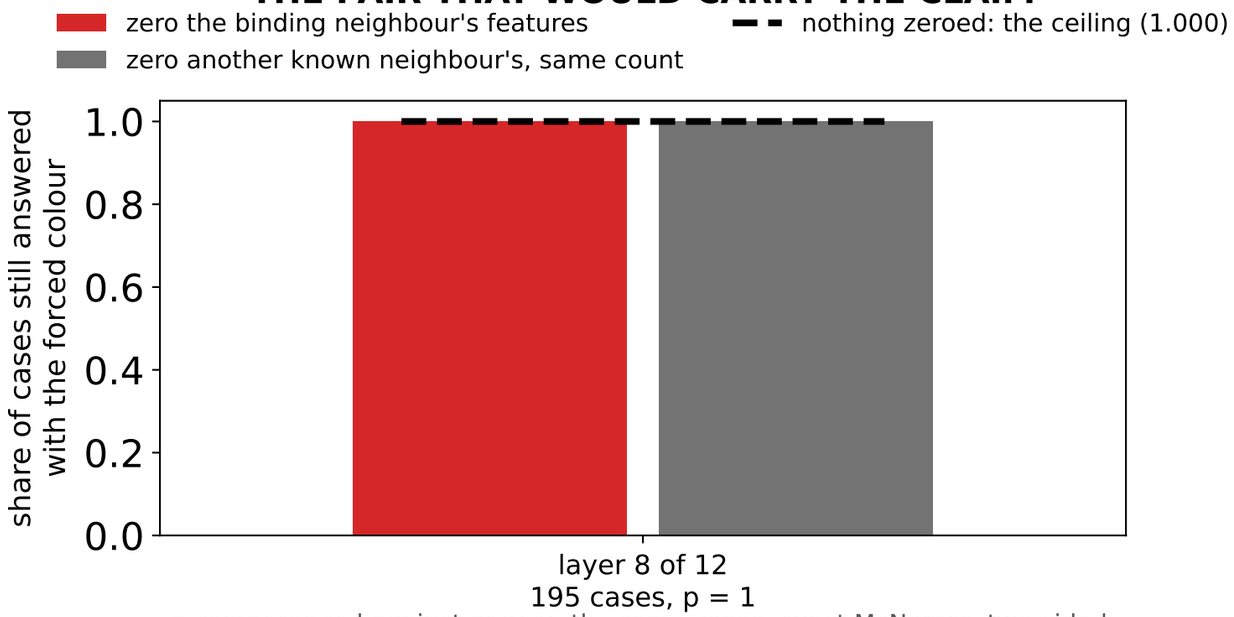
A feature is COUNTED only when it matches one of DD's own predicates at F1 >= 0.30 (the harmonic mean of precision and recall) AND beats a shuffled-label null. Three things to read: (1) the axis predicate, 'a KNOWN neighbour of the node being emitted', IS that node's FORCING SET here (slide 2), and it is encoded where DD's rule needs it -- at the token that predicts the COLOUR, while at the token picking the node nothing clears the 0.30 bar; (2) the SOLVER is NOT more interpretable than the WEAK TWIN, as their equal rule scores already said; (3) THE LEAK-READER ENCODES IT TOO -- 15 features, but only at L10, L12, after the window where the SOLVER uses it. Encoding it is not using it: the evidence is the four intervention pages.

CORRELATIONAL -- this page says WHERE the structure sits, not whether the model uses it. Sparse autoencoder, 4096 features per layer, ~64 active per token, trained on 53,200 activations per layer, then scored separately on the 26,600 positions before a colour token and the 26,600 before a node token, so counts compare within a mask only. Layer 12 is the residual stream after the last block, which is why this axis runs one further than the patching page's. One dictionary per layer per arm, one seed.

Nothing we remove moves the answer -- not the binding neighbour's features, not the whole forcing set, not a random set of the same size. This test has no sensitivity to report from.

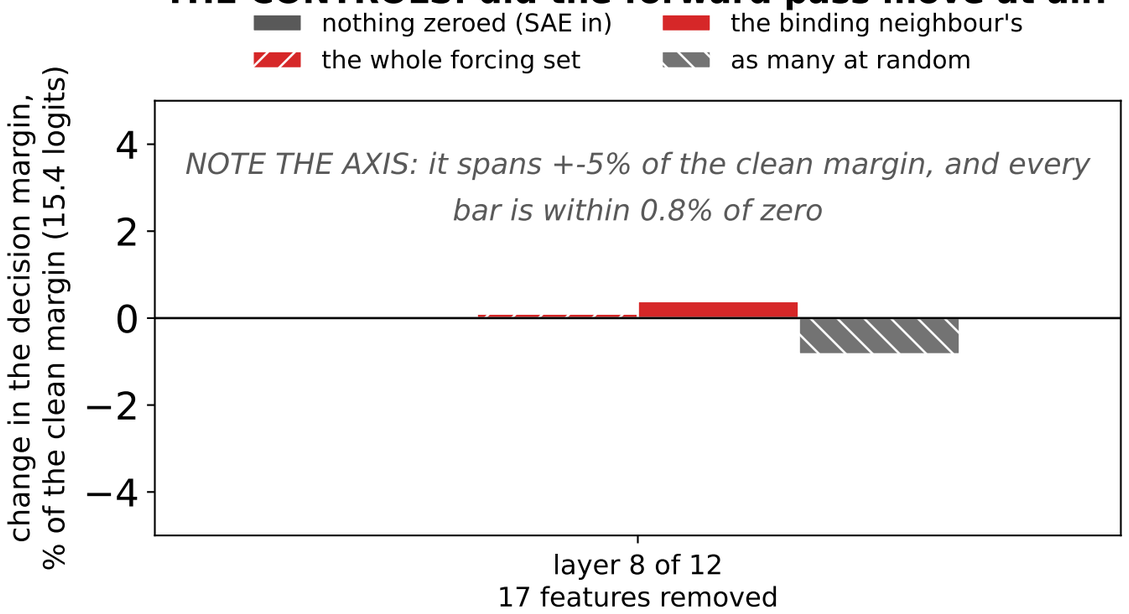
The binding neighbour is the one whose colour rules out the node's last alternative; its fellow known neighbours are just as known, but not decisive.

THE PAIR THAT WOULD CARRY THE CLAIM



p compares red against grey on the same cases, exact McNemar, two-sided.
A case counts only if it owns features for both kinds of neighbour.

THE CONTROLS: did the forward pass move at all?



THE DESIGN: every bar is read against the reconstruction condition (the SAE's own output spliced back in, nothing zeroed), so reconstruction error is netted out rather than charged to the ablation. The two bars on the left remove the same number of features, of the same kind, at the same position, in the same layer, on the same cases; they differ only in which neighbour the features refer to, the one whose colour does the forcing or another known neighbour whose colour does not. Only the model reading that neighbour's colour to eliminate a candidate can open a gap between them. The powered replication -- the same test widened to every feature that beats its null -- runs on the SAME 195 paired cases at layer 8, not a second sample, and gives +0.00, p = 1. The same test at the token that picks the NODE found features to remove there too (35 at layer 8), and removing those changed nothing either.

READ IT AS: inconclusive -- and the control says so before the headline does. Removing the whole forcing set had to cost more than removing a count-matched random set; at layer 8 both cost nothing, 1.000 against a 1.000 baseline, and the paired comparison has 195 cases with not one discordant answer, 95% CI [-0.015, +0.015]. The largest of these ablations moves the decision margin by 0.8% of its clean value, and removing the forcing set moves it the wrong way, towards MORE confidence. So this is not evidence that the features go unused. It is a test in which zeroing 17 of 4096 features at one token cannot perturb a decision this overdetermined -- no sensitivity, so no null to report. The verdict on the next page rests on the interventions, which do move the answer.

Dictionaries: the SOLVER, 4096 features a layer, one seed. The 300 cases are held-out first-round deductions whose forced colour one neighbour's recolouring can flip; features come from the previous page's threshold.

The verdict, and what it is not

Working picture: sweep nodes in index order; emit one only when its known neighbours leave it a single colour, counting the ones it just derived. That is DD's propagation step.

WHAT IT DOES DO (causal unless marked)

* Applies DD's elimination rule: 1.000 on 350 one-token counterfactuals.

* Chains it off its OWN deductions: 0.980, with no decay across the rounds.

* Withholds it: break the forcing and it stops naming the node 1.000 of the time, while a placebo edit leaves it naming v 0.940. But ALL THREE models pass this one, so passing it is not by itself evidence of DD.

* The WEAK TWIN withholds only 0.824 -- the first test here to separate the TWINS, and the same order as its 0.146 solve gap.

* Localised in depth: edited hint -> L5, handoff L6-L10, prediction by L10.

* CORRELATIONAL: sparse features for DD's forcing set peak mid-stack -- but the LEAK-READER carries them too. Ablating them moves nothing; nor do the controls.

WHAT IT DOES NOT DO

* Choose the node ORDER by deduction. DD's own traces are in ascending index order in 100.0% of rounds (3,746 rounds), and a hand rule that does no deduction -- 'next unknown index after the last one' -- scores 0.632 on the metric the model's own node features are scored on (slide 3). WHICH nodes it emits tracks determinacy; the ORDER does not.

* Carry the EDITED FACT through the adjacency rows: patching them recovers at most 0.01, in any layer, in any of the three models. Reading that table is a separate matter -- see below.

* Concentrate the rule anywhere we can point at: ablating the worst of the 48 heads leaves 84% of it. Distributed; this deck says THAT it runs DD.

* The LEAK-READER barely learned the rule: 0.388 [0.33, 0.45], above the 0.250 chance rate but far below the SOLVER. Shown an edited deduction it keeps its own answer 99.2% of the time: it reads the adjacency-order leak.

LIMITATIONS. The interventions show the model propagating constraints the way DD does; they do NOT pin it to DD's exact algorithm, and a different propagator with the same one-step behaviour would pass every test on this deck. One checkpoint per model and no seed replicates, so every leak-free vs leaked contrast is also two separate training runs. One task configuration (N = 24, degree 5, q = 4), so nothing here is a claim about depth or scale, and the per-depth bins run as thin as a few dozen cases. The withhold gap and the solve-rate gap are different measurements on different samples: their agreement is a consistency check, not a decomposition. The sparse-autoencoder pages are correlational, and the ablation that would make them causal moved no answer at all, controls included -- inconclusive, not negative.